# Credit Card Risk Prediction

FRA Project - Milestone 1

BY
DIVYA KURUVADA

# CONTENT



1. Problem Statement
2. Data Overview
3. Data Cleaning & Processing
4. Data Analysis
5. Model Building
6. Prediction & Validation

Businesses or companies can fall prey to default if they are not able to keep up their debt obligations. Defaults will lead to a lower credit rating for the company which in turn reduces its chances of getting credit in the future and may have to pay higher interests on existing debts as well as any new obligations. From an investor's point of view, he would want to invest in a company if it is capable of handling its financial obligations, can grow quickly, and is able to manage the growth scale.

A balance sheet is a financial statement of a company that provides a snapshot of what a company owns, owes, and the amount invested by the shareholders. Thus, it is an important tool that helps evaluate the performance of a business.

Data that is available includes information from the financial statement of the companies for the previous year (2015). Also, information about the Networth of the company in the following year (2016) is provided which can be used to drive the labelled field.

Explanation of data fields available in Data Dictionary, 'Credit Default Data Dictionary.xlsx'

# 2.1 Glimpse of Dataset

| | Co_Code | Co_Name | Networth Next Year | Equity Paid Up | Networth | Capital Employed | Total Debt | Gross Block | Net Working Capital | Current Assets | Current Liabilities and Provisions | Total Assets/Liabilities | Gross Sales | Net Sales | Other Income |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 16974 | Hind.Cables | -8021.60 | 419.36 | -7027.48 | -1007.24 | 5936.03 | 474.30 | -1076.34 | 40.50 | 1116.85 | 109.60 | 0.00 | 0.00 | 7.60 |
| 1 | 21214 | Tata Tele. Mah. | -3986.19 | 1954.93 | -2968.08 | 4458.20 | 7410.18 | 9070.86 | -1098.88 | 486.86 | 1585.74 | 6043.94 | 2892.73 | 2892.73 | 46.27 |
| 2 | 14852 | ABG Shipyard | -3192.58 | 53.84 | 506.86 | 7714.68 | 6944.54 | 1281.54 | 4496.25 | 9097.64 | 4601.39 | 12316.07 | 392.13 | 392.13 | 9.55 |
| 3 | 2439 | GTL | -3054.51 | 157.30 | -623.49 | 2353.88 | 2326.05 | 1033.69 | -2612.42 | 1034.12 | 3646.54 | 6000.42 | 1354.39 | 1354.39 | 223.85 |
| 4 | 23505 | Bharati Defence | -2967.36 | 50.30 | -1070.83 | 4675.33 | 5740.90 | 1084.20 | 1836.23 | 4685.81 | 2849.58 | 7524.91 | 38.72 | 38.72 | 9.82 |

| Value Of Output | Cost of Production | Selling Cost | PBIDT | PBDT | PBIT | PBT | PAT | Adjusted PAT | CP | Revenue earnings in forex | Revenue expenses in forex | Capital expenses in forex | Book Value (Unit Curr) | Book Value (Adj.) (Unit Curr) | Market Capitalisation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -0.07 | 137.67 | 0.00 | -179.06 | -926.52 | -185.53 | -932.99 | -932.99 | -937.85 | -926.52 | 0.00 | 0.00 | 0.00 | -167.58 | -167.58 | 0.00 |
| 2900.71 | 2572.46 | 40.51 | 646.46 | -4.32 | 35.53 | -615.25 | -615.25 | -617.14 | -4.32 | 6.35 | 143.42 | 141.17 | -15.18 | -15.18 | 1544.39 |
| 301.16 | 408.51 | 54.83 | -281.92 | -1086.71 | -381.10 | -1185.89 | -897.70 | -873.39 | -798.52 | 0.00 | 86.36 | 2.27 | 94.14 | 94.14 | 1220.81 |
| 1350.14 | 1326.99 | 3.34 | -213.01 | -677.57 | -336.73 | -801.29 | -801.29 | -770.18 | -677.57 | 0.89 | 28.88 | 0.00 | -39.64 | -39.64 | 194.27 |
| 38.72 | 186.29 | 1.97 | -647.86 | -944.42 | -710.13 | -1006.69 | -864.58 | -327.77 | -802.31 | 0.00 | 15.62 | 0.00 | -212.89 | -212.89 | 113.68 |

| CEPS (annualised) (Unit Curr) | Cash Flow From Operating Activities | Cash Flow From Investing Activities | Cash Flow From Financing Activities | ROG-Net Worth (%) | ROG-Capital Employed (%) | ROG-Gross Block (%) | ROG-Gross Sales (%) | ROG-Net Sales (%) | ROG-Cost of Production (%) | ROG-Total Assets (%) | ROG-PBIDT (%) | ROG-PBDT (%) | ROG-PBIT (%) | ROG-PBT (%) | ROG-PAT (%) | ROG-CP (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -22.09 | -102.47 | 1.46 | 92.58 | -15.31 | -20.76 | -9.75 | 0.00 | 0.00 | 1.38 | -25.75 | 5.23 | -19.29 | 4.43 | -19.33 | -19.33 | -19.29 |
| -0.02 | 635.91 | -785.00 | 176.93 | -26.15 | 6.30 | 3.17 | 5.92 | 5.92 | 13.16 | 6.66 | 5.24 | -108.77 | 614.89 | -9.85 | -9.85 | -108.77 |
| -148.31 | -873.40 | -458.27 | 1187.51 | -61.86 | 15.66 | -2.07 | -75.87 | -75.87 | -69.93 | -0.78 | -169.69 | -431.68 | -221.85 | -300.41 | -350.43 | -642.67 |
| -43.08 | 324.47 | 17.31 | -412.55 | -450.67 | -40.84 | -1.80 | 8.33 | 8.33 | 16.59 | -10.12 | -203.84 | -102.72 | -455.28 | -80.23 | -70.57 | -88.52 |
| -159.50 | -191.54 | 61.34 | 143.65 | -559.83 | -11.76 | 0.34 | -80.61 | -80.61 | -48.80 | -5.81 | -40.07 | 3.28 | -38.72 | 1.87 | -2.59 | -1.13 |

# 2.1 Glimpse of Dataset

| ROG-Revenue earnings in forex (%) | ROG-Revenue expenses in forex (%) | ROG-Market Capitalisation (%) | Current Ratio[Latest] | Fixed Assets Ratio[Latest] | Inventory Ratio[Latest] | Debtors Ratio[Latest] | Total Asset Turnover Ratio[Latest] | Interest Cover Ratio[Latest] | PBIDTM (%) [Latest] | PBITM (%) [Latest] | PBDTM (%) [Latest] | CPM (%) [Latest] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 0.00 | 0.00 | 0.02 | 0.00 | 0.00 | 0.00 | 0.00 | -0.20 | 0.00 | 0.00 | 0.00 | 0.00 |
| 1.93 | 34.00 | 6.61 | 0.08 | 0.25 | 804.44 | 10.35 | 0.30 | -0.84 | -10.30 | -39.74 | -57.74 | -57.74 |
| -100.00 | -81.21 | -6.31 | 1.06 | 0.03 | 0.01 | 0.42 | 0.00 | -2.21 | -5279.14 | -5516.98 | -7780.25 | -7723.67 |
| -52.91 | -33.85 | -13.94 | 0.09 | 5.08 | 411.15 | 9.26 | 0.00 | -0.16 | -3.33 | -7.21 | -48.13 | -47.70 |
| -100.00 | -91.17 | -12.91 | 0.50 | 0.05 | 0.02 | 1.28 | 0.01 | -0.73 | -295.55 | -400.55 | -845.88 | 379.79 |

| APATM (%) [Latest] | Debtors Velocity (Days) | Creditors Velocity (Days) | Inventory Velocity (Days) | Value of Output/Total Assets | Value of Output/Gross Block |
|---|---|---|---|---|---|
| 0.00 | 0 | 0 | 45.0 | 0.00 | 0.00 |
| -87.18 | 29 | 101 | 2.0 | 0.31 | 0.24 |
| -7961.51 | 97 | 558 | 0.0 | -0.03 | -0.26 |
| -51.58 | 93 | 63 | 2.0 | 0.24 | 1.90 |
| 274.79 | 3887 | 346 | 0.0 | 0.01 | 0.05 |

# 2.2 Data Dictionary

| Col.No | Field Name | Description |
|---|---|---|
| *1* | *Co_Code* | *Company Code* |
| *2* | *Co_Name* | *Company Name* |
| *3* | *Networth Next Year* | *Value of a company as on 2016 - Next Year(difference between the value of total assets and total liabilities)* |
| *4* | *Equity Paid Up* | *Amount that has been received by the company through the issue of shares to the shareholders* |
| *5* | *Networth* | *Value of a company as on 2015 - Current Year* |
| *6* | *Capital Employed* | *Total amount of capital used for the acquisition of profits by a company* |
| *7* | *Total Debt* | *The sum of money borrowed by the company and is due to be paid* |
| *8* | *Gross Block* | *Total value of all of the assets that a company owns* |
| *9* | *Net Working Capital* | *The difference between a company's current assets (cash, accounts receivable, inventories of raw materials and finished goods) and its current liabilities (accounts payable).* |
| *10* | *Current Assets* | *All the assets of a company that are expected to be sold or used as a result of standard business operations over the next year.* |
| *11* | *Current Liabilities and Provisions* | *Short-term financial obligations that are due within one year (includes amount that is set aside cover a future liability)* |
| *12* | *Total Assets/Liabilities* | *Ratio of total assets to liabailities of the company* |
| *13* | *Gross Sales* | *The grand total of sale transactions within the accounting period* |
| *14* | *Net Sales* | *Gross sales minus returns, allowances, and discounts* |

# 2.2 Data Dictionary

| Col.No | Field Name | Description |
|---|---|---|
| 15 | Other Income | Income realized from non-business activities (e.g. sale of long term asset) |
| 16 | Value Of Output | Product of physical output of goods and services produced by company and its market price |
| 17 | Cost of Production | Costs incurred by a business from manufacturing a product or providing a service |
| 18 | Selling Cost | Costs which are made to create the demand for the product (advertising expenditures, packaging and styling, salaries, commissions and travelling expenses of sales personnel, and the cost of shops and showrooms) |
| 19 | PBIDT | Profit Before Interest, Depreciation &Taxes |
| 20 | PBDT | Profit Before Depreciation and Tax |
| 21 | PBIT | Profit before interest and taxes |
| 22 | PBT | Profit before tax |
| 23 | PAT | Profit After Tax |
| 24 | Adjusted PAT | Adjusted profit is the best estimate of the true profit |
| 26 | CP | Commercial paper , a short-term debt instrument to meet short-term liabilities. |
| 27 | Revenue earnings in forex | Revenue earned in foreign currency |
| 28 | Revenue expenses in forex | Expenses due to foreign currency transactions |
| 29 | Capital expenses in forex | Long term investment in forex |
| 30 | Book Value (Unit Curr) | Net asset value |
| 31 | Book Value (Adj.) (Unit Curr) | Book value adjusted to reflect asset's true fair market value |

# 2.2 Data Dictionary

| Col.No | Field Name | Description |
|---|---|---|
| 32 | Market Capitalisation | Product of the total number of a company's outstanding shares and the current market price of one share |
| 33 | CEPS (annualised) (Unit Curr) | Cash Earnings per Share, profitability ratio that measures the financial performance of a company by calculating cash flows on a per share basis |
| 34 | Cash Flow From Operating Activities | Use of cash from ongoing regular business activities |
| 35 | Cash Flow From Investing Activities | Cash used in the purchase of non-current assets–or long-term assets– that will deliver value in the future |
| 36 | Cash Flow From Financing Activities | Net flows of cash that are used to fund the company (transactions involving debt, equity, and dividends) |
| 37 | ROG-Net Worth (%) | Rate of Growth - Networth |
| 38 | ROG-Capital Employed (%) | Rate of Growth - Capital Employed |
| 39 | ROG-Gross Block (%) | Rate of Growth - Gross Block |
| 40 | ROG-Gross Sales (%) | Rate of Growth - Gross Sales |
| 41 | ROG-Net Sales (%) | Rate of Growth - Net Sales |
| 42 | ROG-Cost of Production (%) | Rate of Growth - Cost of Production |
| 43 | ROG-Total Assets (%) | Rate of Growth - Total Assets |
| 44 | ROG-PBIDT (%) | Rate of Growth- PBIDT |
| 45 | ROG-PBDT (%) | Rate of Growth- PBDT |
| 46 | ROG-PBIT (%) | Rate of Growth- PBIT |
| 47 | ROG-PBT (%) | Rate of Growth- PBT |

## 2.2 Data Dictionary

| Col.No | Field Name | Description |
|---|---|---|
| 48 | ROG-PAT (%) | Rate of Growth- PAT |
| 49 | ROG-CP (%) | Rate of Growth- CP |
| 50 | ROG-Revenue earnings in forex (%) | Rate of Growth - Revenue earnings in forex |
| 51 | ROG-Revenue expenses in forex (%) | Rate of Growth - Revenue expenses in forex |
| 52 | ROG-Market Capitalisation (%) | Rate of Growth - Market Capitalisation |
| 53 | Current Ratio[Latest] | Liquidity ratio, company's ability to pay short-term obligations or those due within one year |
| 54 | Fixed Assets Ratio[Latest] | Solvency ratio, the capacity of a company to discharge its obligations towards long-term lenders indicating |
| 55 | Inventory Ratio[Latest] | Activity ratio, specifies the number of times the stock or inventory has been replaced and sold by the company |
| 56 | Debtors Ratio[Latest] | Measures how quickly cash debtors are paying back to the company |
| 57 | Total Asset Turnover Ratio[Latest] | The value of a company's revenues relative to the value of its assets |
| 58 | Interest Cover Ratio[Latest] | Determines how easily a company can pay interest on its outstanding debt |
| 59 | PBIDTM (%)[Latest] | Profit before Interest Depreciation and Tax Margin |
| 60 | PBITM (%)[Latest] | Profit Before Interest Tax Margin |
| 61 | PBDTM (%)[Latest] | Profit Before Depreciation Tax Margin |
| 62 | CPM (%)[Latest] | Cost per thousand (advertising cost) |
| 63 | APATM (%)[Latest] | After tax profit margin |

## 2.2 Data Dictionary

| Col.No | Field Name | Description |
|---|---|---|
| *64* | *Debtors Velocity (Days)* | *Average days required for receiving the payments* |
| *65* | *Creditors Velocity (Days)* | *Average number of days company takes to pay suppliers* |
| *66* | *Inventory Velocity (Days)* | *Average number of days the company needs to turn its inventory into sales* |
| *67* | *Value of Output/Total Assets* | *Ratio of Value of Output (market value) to Total Assets* |
| *68* | *Value of Output/Gross Block* | *Ratio of Value of Output (market value) to Gross Block* |

There are total 68 variables in this dataset.

It contains various measures related to company business.

## 2.2 Data info

The number of rows (observations) is 3586.
The number of columns (variables) is 67.

There are 63 float variables,3 integer variables and 1 object variable.

There are no duplicates present in the dataset.

All the data has been renamed for ease as suggested in original dictionary dataset.

Networth_next_Year will be taken as ‘default’ column or

output variable.

We are dropping the company code and

## 2.3 Statistical Summary

| | Co_Code | Networth Next Year | Equity Paid Up | Networth | Capital Employed | Total Debt | Gross Block | Net Working Capital | Current Assets | Current Liabilities and Provisions | Total Assets/Liabilities | Gross Sales | Net Sales |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| count | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 | 3586.00 |
| mean | 16065.39 | 725.05 | 62.97 | 649.75 | 2799.61 | 1994.82 | 594.18 | 410.81 | 1960.35 | 391.99 | 1778.45 | 1123.74 | 1079.70 |
| std | 19776.82 | 4769.68 | 778.76 | 4091.99 | 26975.14 | 23652.84 | 4871.55 | 6301.22 | 22577.57 | 2675.00 | 11437.57 | 10603.70 | 9996.57 |
| min | 4.00 | -8021.60 | 0.00 | -7027.48 | -1824.75 | -0.72 | -41.19 | -13162.42 | -0.91 | -0.23 | -4.51 | -62.59 | -62.59 |
| 25% | 3029.25 | 3.98 | 3.75 | 3.89 | 7.60 | 0.03 | 0.57 | 0.94 | 4.00 | 0.73 | 10.56 | 1.44 | 1.44 |
| 50% | 6077.50 | 19.02 | 8.29 | 18.58 | 39.09 | 7.49 | 15.87 | 10.14 | 24.54 | 9.23 | 52.01 | 31.21 | 30.44 |
| 75% | 24269.50 | 123.80 | 19.52 | 117.30 | 226.60 | 72.35 | 131.90 | 61.18 | 135.28 | 65.65 | 310.54 | 242.25 | 234.44 |
| max | 72493.00 | 111729.10 | 42263.46 | 81657.35 | 714001.25 | 652823.81 | 128477.59 | 223257.56 | 721166.00 | 83232.98 | 254737.22 | 474182.94 | 443775.16 |

The statistical summary shows that there are many outliers present in the dataset in almost all the variables.

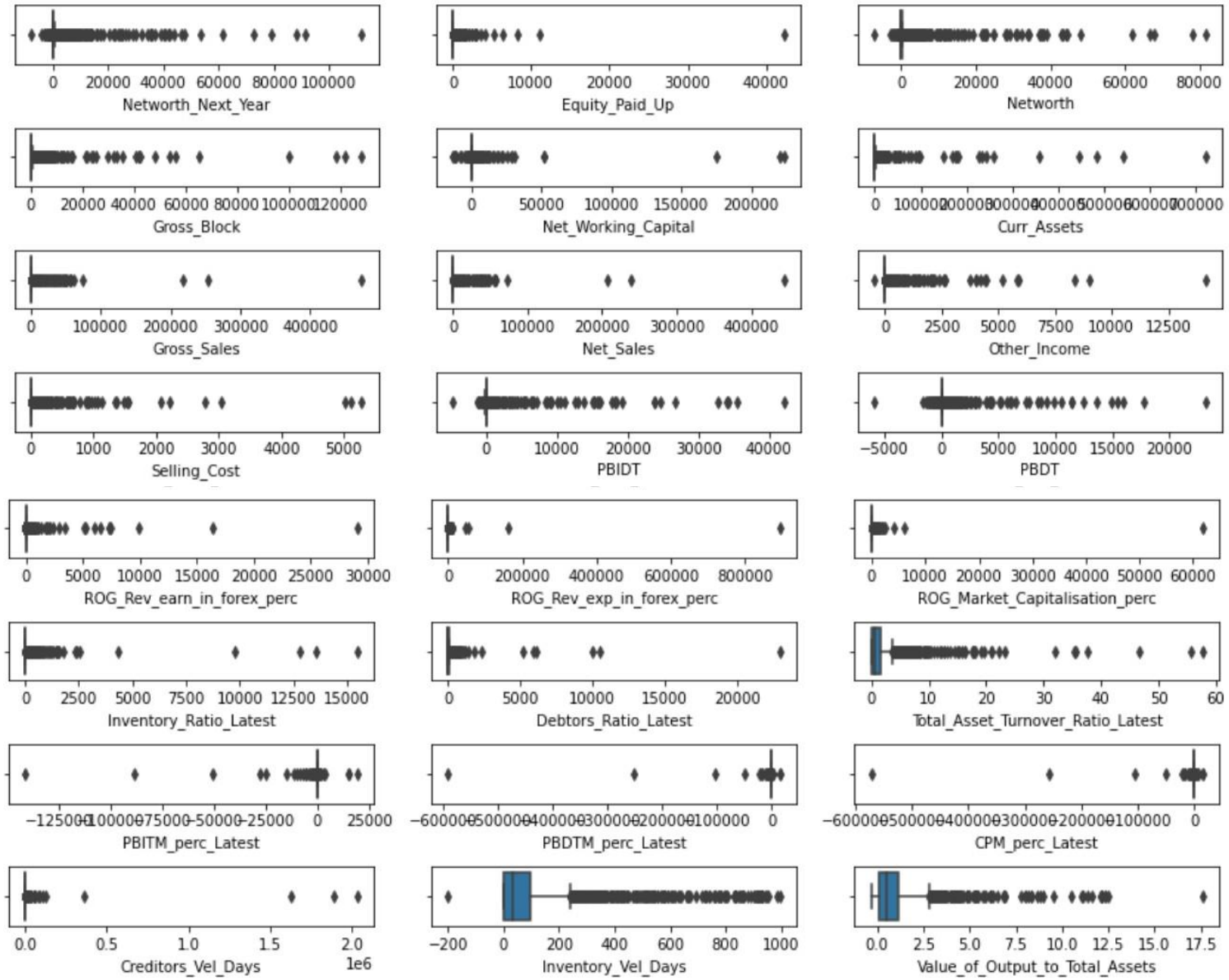
The median Networth is 18.58 units where as minimum is -7027 units and maximum is 81657 units. This shows that how much deviation is there which is also proved by the difference in mean and std. deviation.

ROG-Revenue earnings in forex and expense variable shows that the $25^{th}$ & $75^{th}$ percentile are 0 which means they have a large chunk of data having zero which might not be contributing enough towards the output and can be eliminated.

We will look further into the outliers to see how bad the situation is, whether any treatment is required or not.

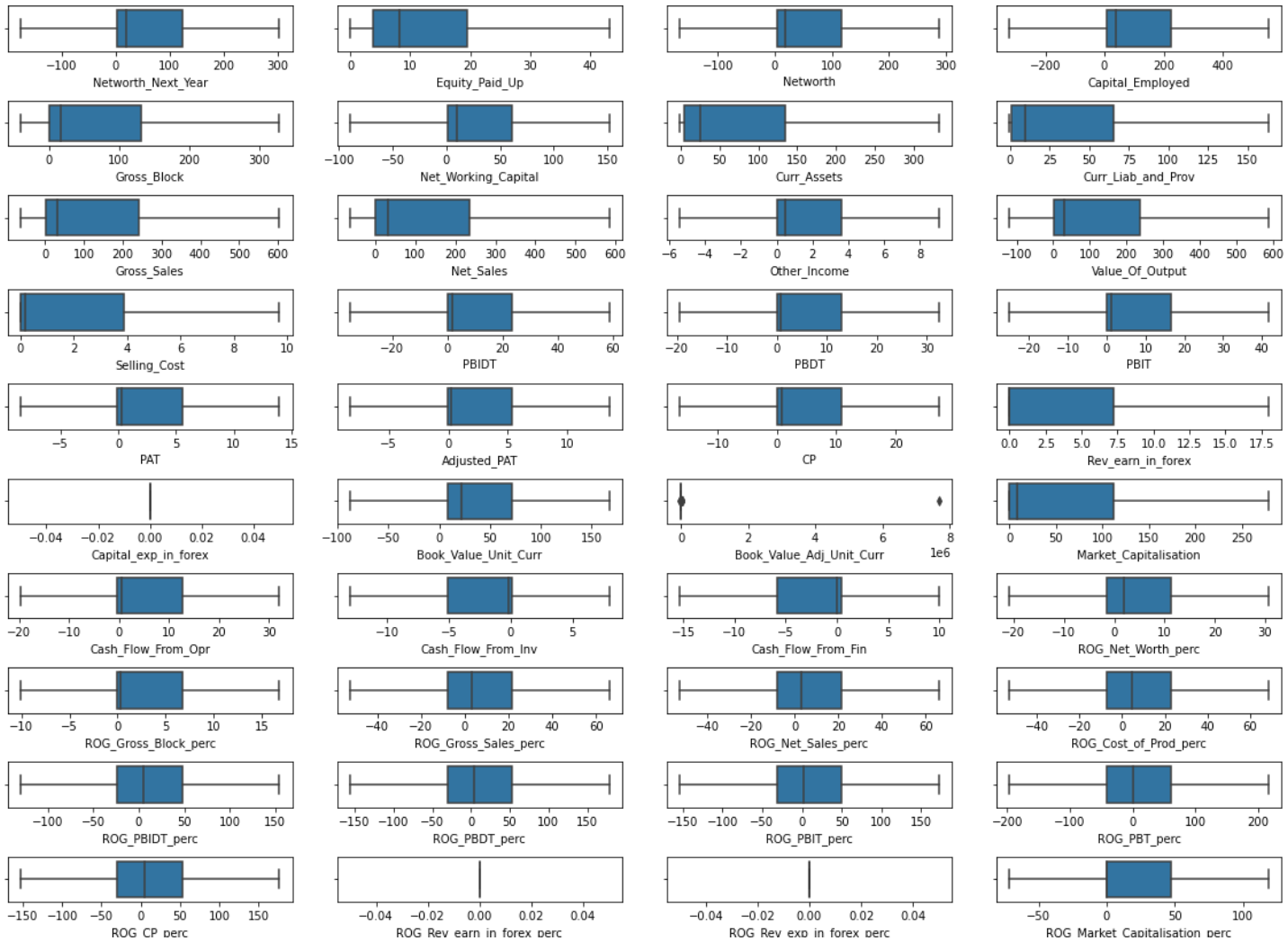
# Box Plots showing Outliers

## 3.1 Outlier Treatment



# After Outlier Treatment

## 3.1 Outlier Treatment



## 3.2 Transform Target Variable

Target value 'Netwoth_Next_Year' is transform into 0 and 1 s.

Netwoth_Next_Year <0 (negative) then target variable = 1
Netwoth_Next_Year >=0 (positive) then target variable = 0

1 means 'Company might Default'
0 means 'Company might not Default'

| Debtors_Vel_Days | Creditors_Vel_Days | Inventory_Vel_Days | Value_of_Output_to_Total_Assets | Value_of_Output_to_Gross_Block | default |
|---|---|---|---|---|---|
| 0.0 | 0.0 | 45.0 | 0.00 | 0.00 | 1 |
| 29.0 | 101.0 | 2.0 | 0.31 | 0.24 | 1 |
| 97.0 | 210.5 | 0.0 | -0.03 | -0.26 | 1 |
| 93.0 | 63.0 | 2.0 | 0.24 | 1.90 | 1 |
| 253.0 | 210.5 | 0.0 | 0.01 | 0.05 | 1 |

Totals number of 0's are 3198 and total number of 1's are 388.

## 3.3 Missing Value Treatment

There are 118 missing values present in the dataset.

| | |
|---|---|
| Inventory_Vel_Days | 103 |
| Book_Value_Adj_Unit_Curr | 4 |
| PBITM_perc_Latest | 1 |
| Curr_Ratio_Latest | 1 |
| Fixed_Assets_Ratio_Latest | 1 |
| Inventory_Ratio_Latest | 1 |
| Debtors_Ratio_Latest | 1 |
| Interest_Cover_Ratio_Latest | 1 |
| PBIDTM_perc_Latest | 1 |
| Total_Asset_Turnover_Ratio_Latest | 1 |
| PBDTM_perc_Latest | 1 |
| CPM_perc_Latest | 1 |
| APATM_perc_Latest | 1 |

missing values in percentage

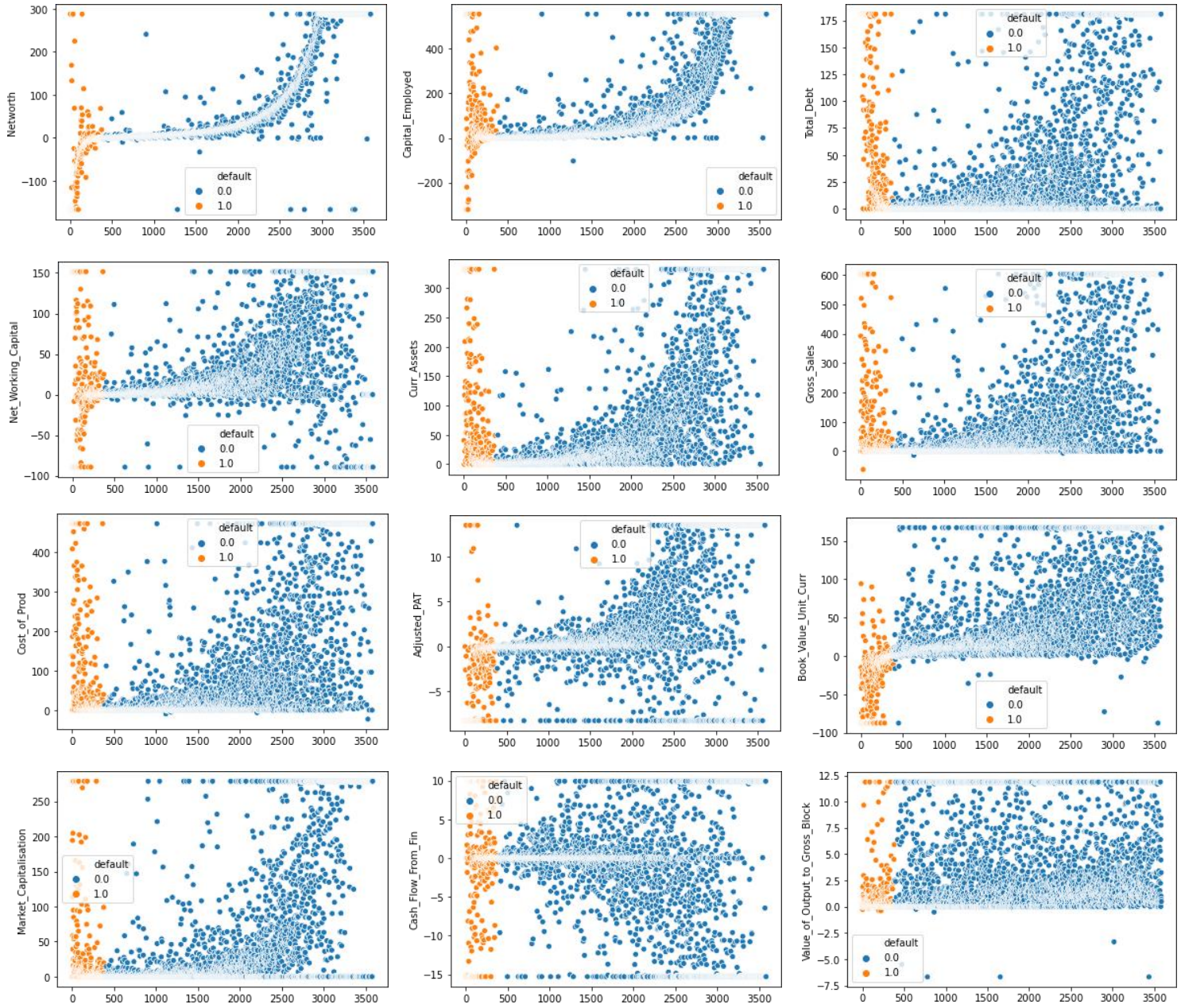
| | |
|---|---|
| Inventory_Vel_Days | 2.872281 |
| Book_Value_Adj_Unit_Curr | 0.111545 |
| PBITM_perc_Latest | 0.027886 |
| Curr_Ratio_Latest | 0.027886 |
| Fixed_Assets_Ratio_Latest | 0.027886 |
| Inventory_Ratio_Latest | 0.027886 |
| Debtors_Ratio_Latest | 0.027886 |
| Interest_Cover_Ratio_Latest | 0.027886 |
| PBIDTM_perc_Latest | 0.027886 |
| Total_Asset_Turnover_Ratio_Latest | 0.027886 |
| PBDTM_perc_Latest | 0.027886 |
| CPM_perc_Latest | 0.027886 |
| APATM_perc_Latest | 0.027886 |

The missing value is quite less in percentage, hence shouldn't be removed, rather imputed.

| | |
|---|---|
| Inventory_Vel_Days | 0 |
| Book_Value_Adj_Unit_Curr | 0 |
| PBITM_perc_Latest | 0 |
| Curr_Ratio_Latest | 0 |
| Fixed_Assets_Ratio_Latest | 0 |
| Inventory_Ratio_Latest | 0 |
| Debtors_Ratio_Latest | 0 |
| Interest_Cover_Ratio_Latest | 0 |
| PBIDTM_perc_Latest | 0 |
| Total_Asset_Turnover_Ratio_Latest | 0 |
| PBDTM_perc_Latest | 0 |
| CPM_perc_Latest | 0 |
| APATM_perc_Latest | 0 |

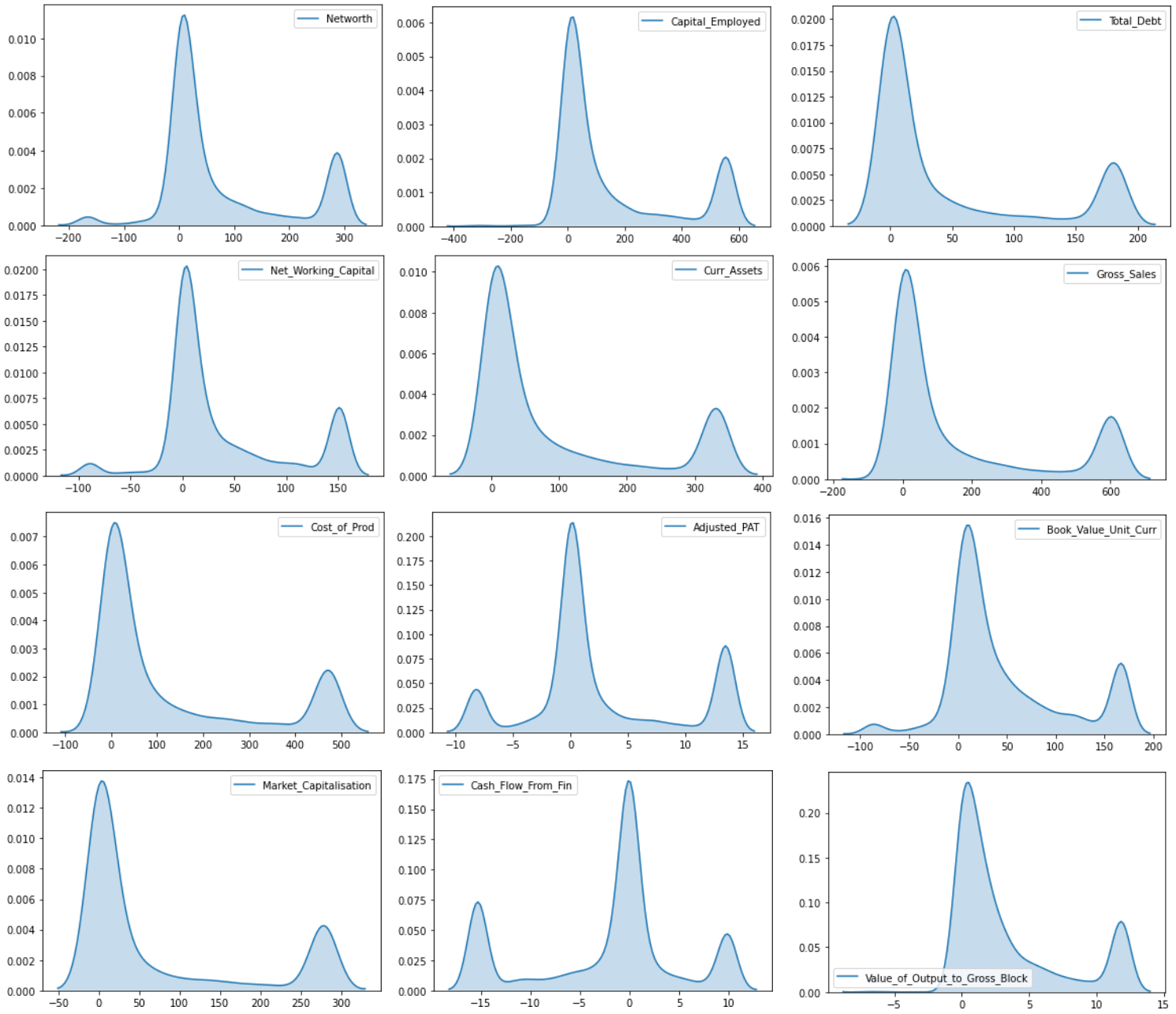
# 4.1 Univariate Analysis

# Scatter Plot of few Variables



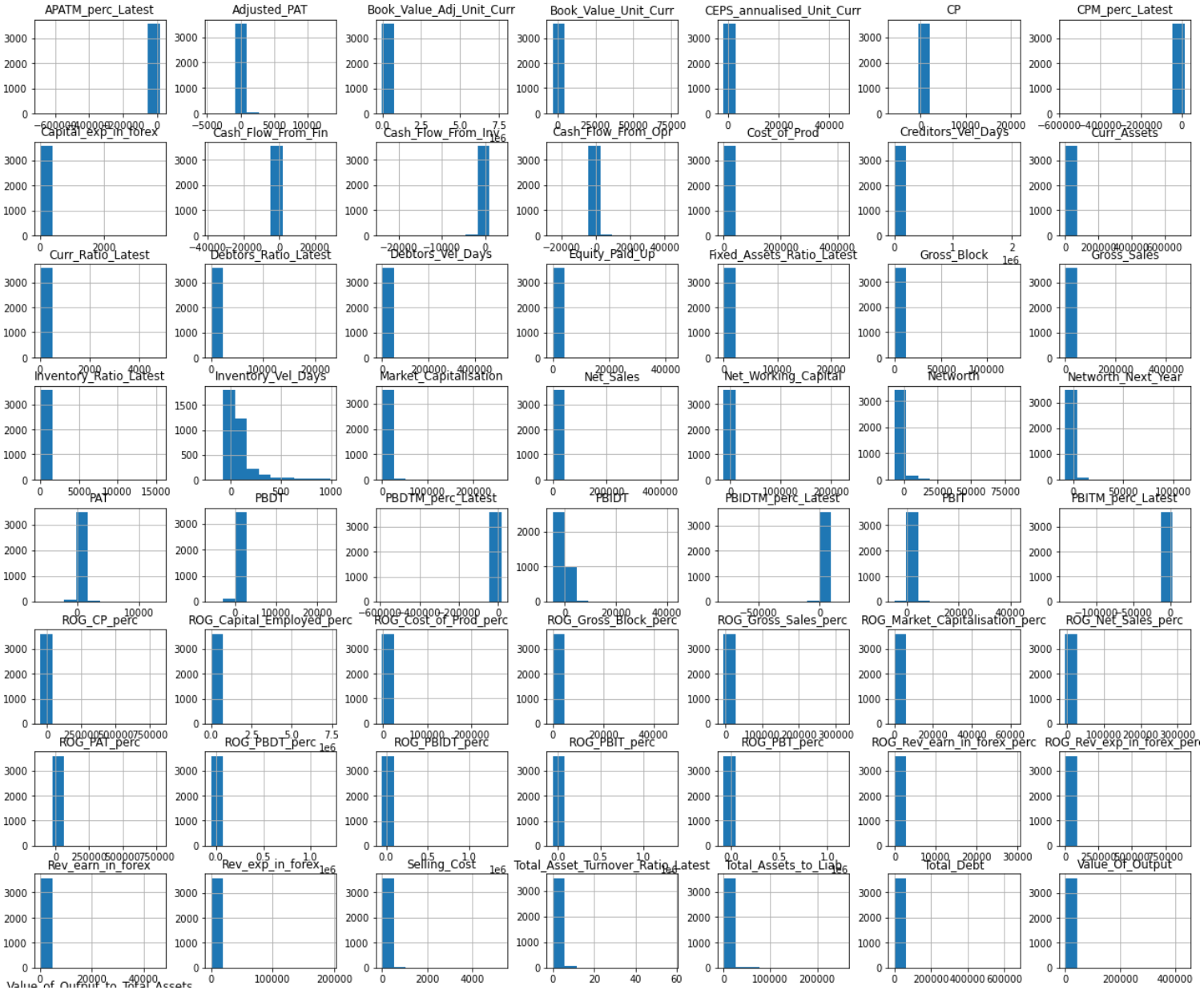
# 4.1 Univariate Analysis

# Density Plot of few Variables

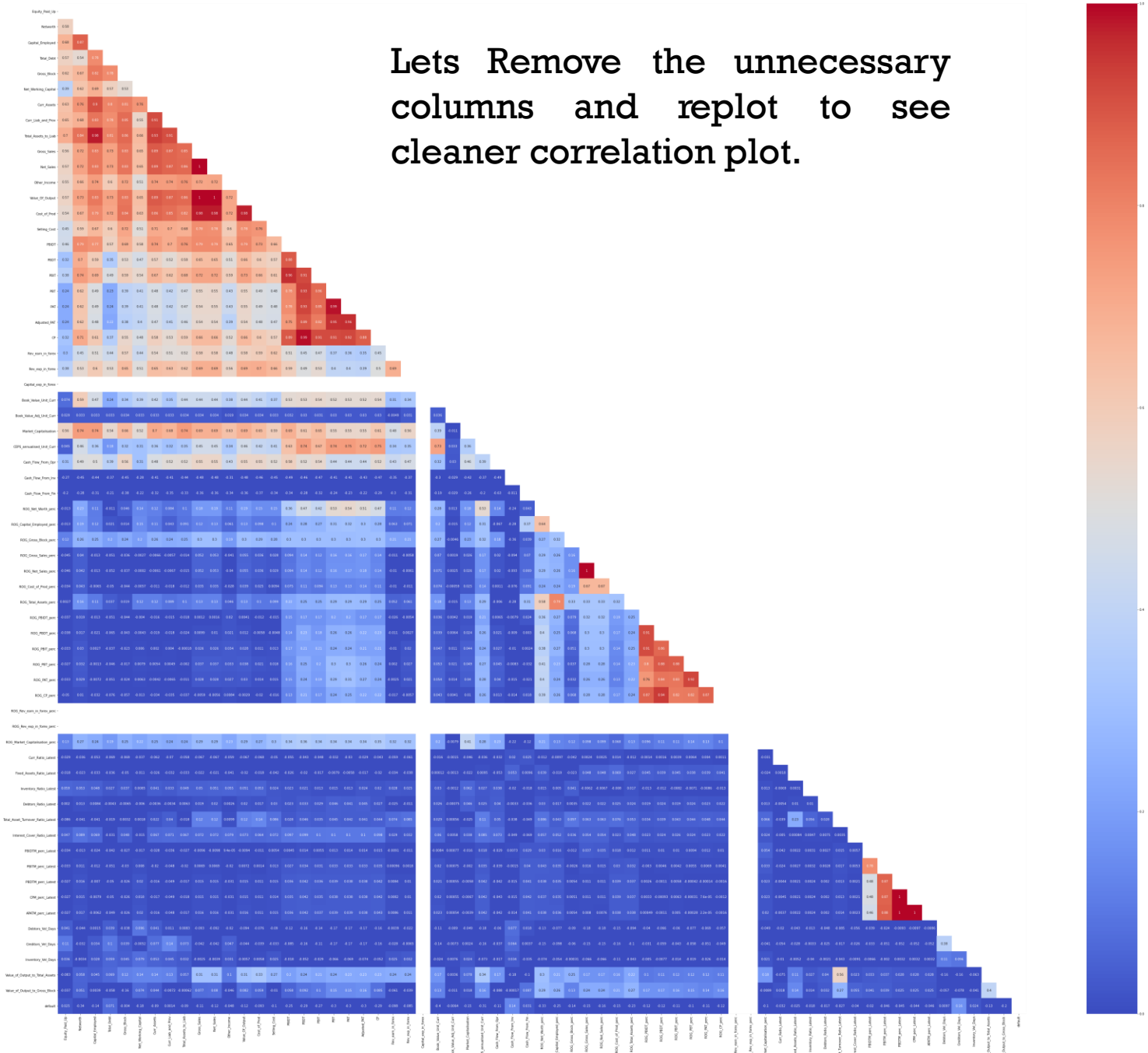


# 4.1 Univariate Analysis

# Histogram

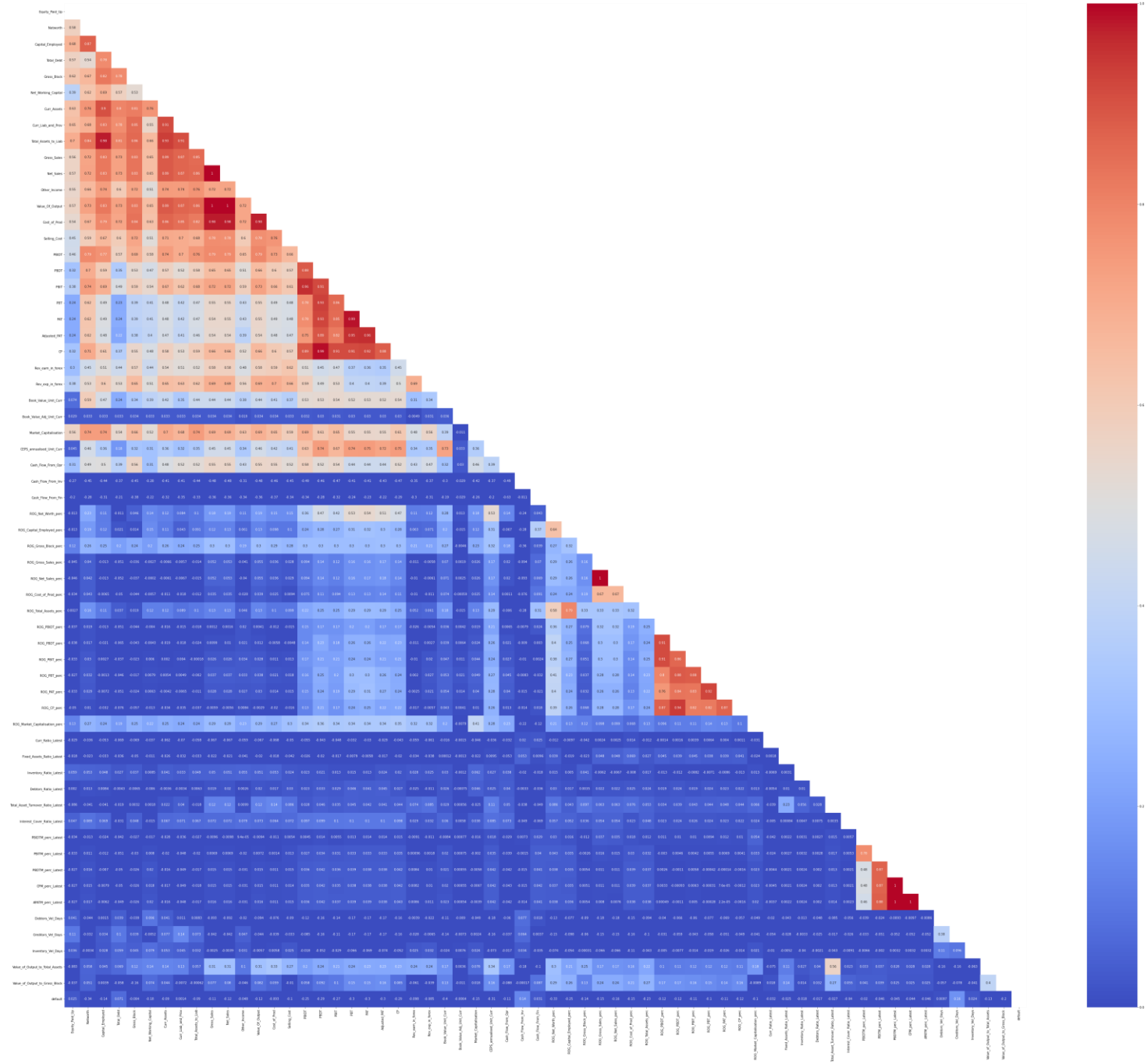


# 4.2 Correlation Plot

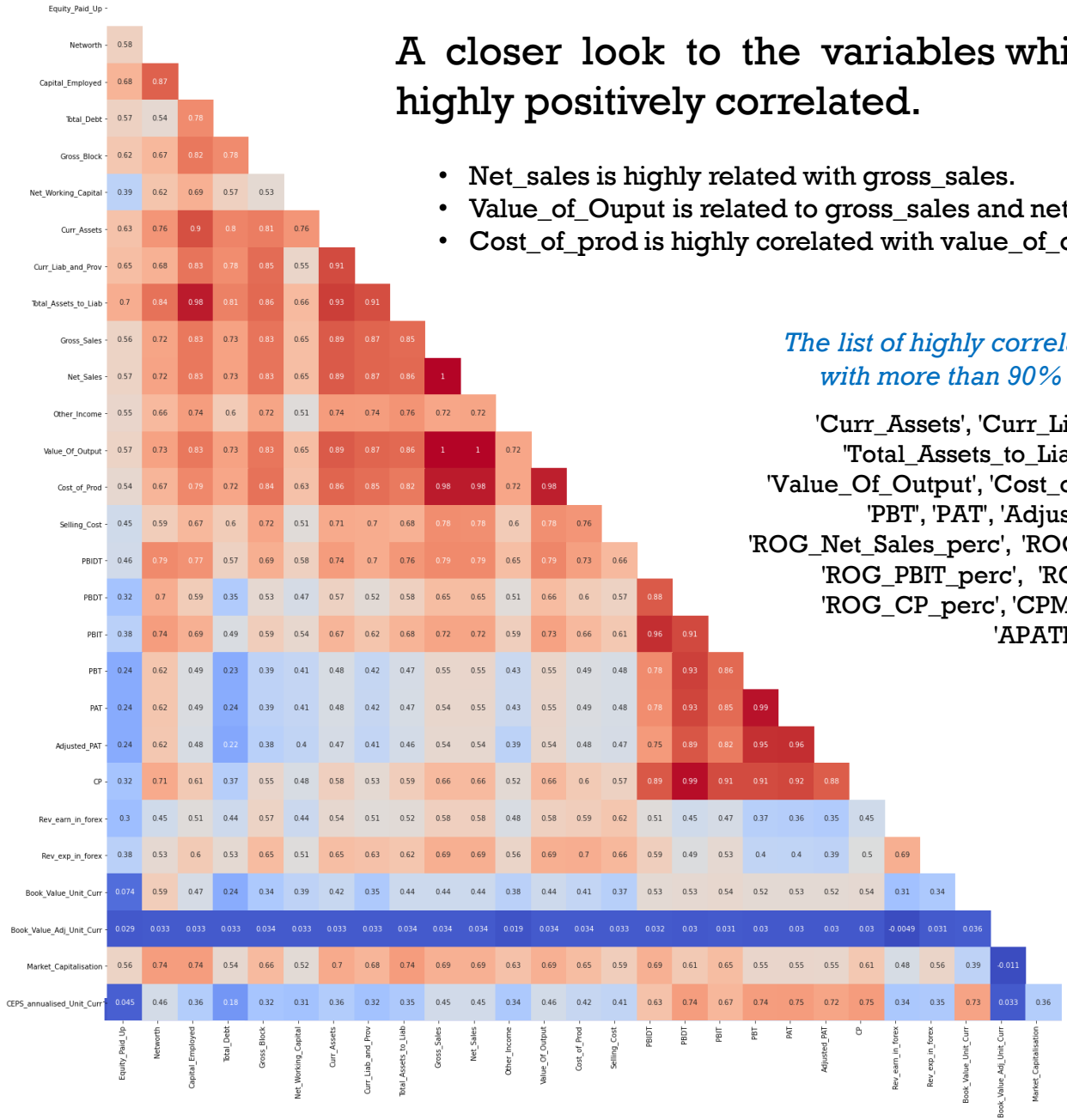


Lets Remove the unnecessary columns and replot to see cleaner correlation plot.

# 4.2 Correlation Plot



# 4.2 Correlation Plot



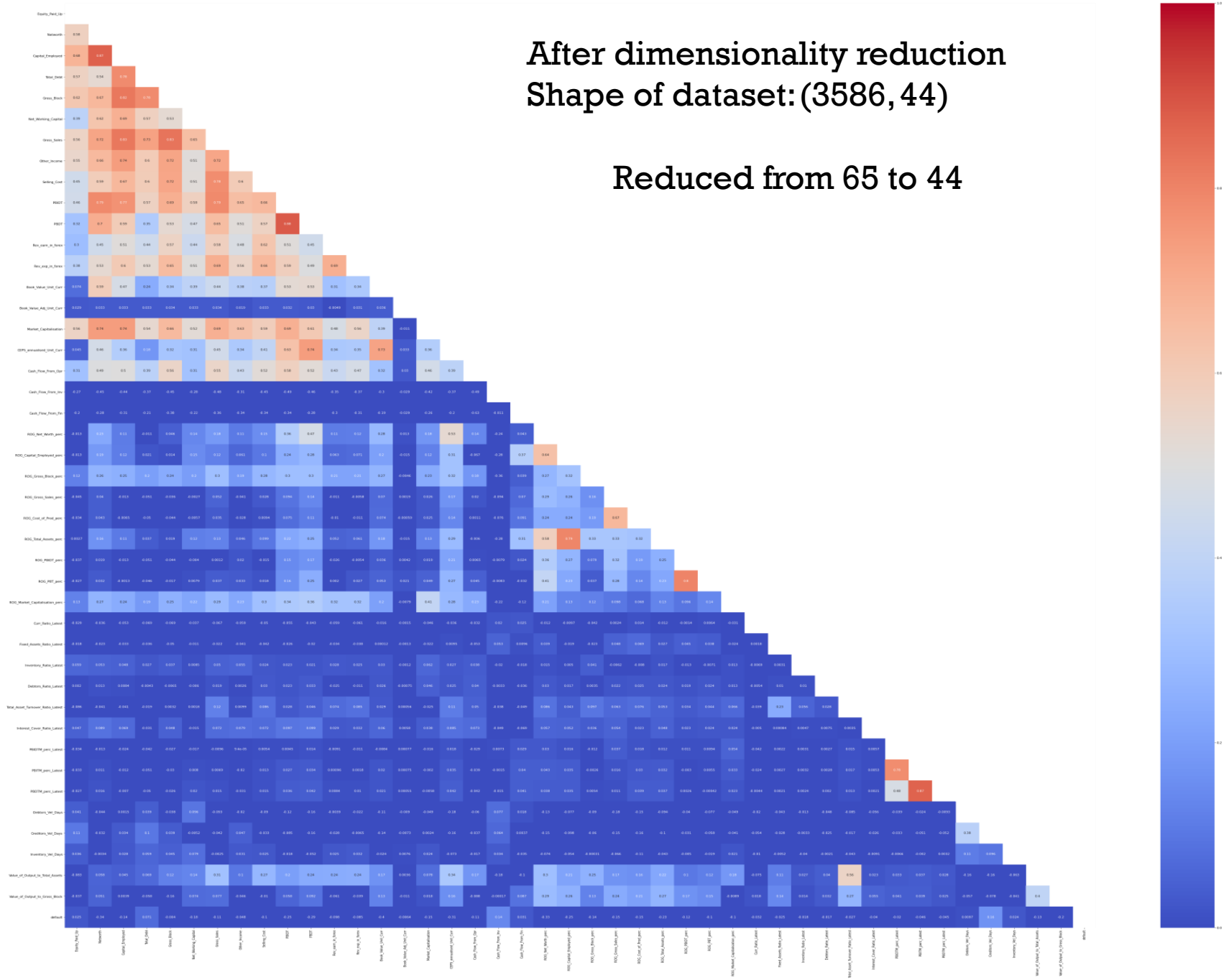

## A closer look to the variables which are highly positively correlated.

- Net_sales is highly related with gross_sales.
- Value_of_Ouput is related to gross_sales and net_sales
- Cost_of_prod is highly corelated with value_of_output.

*The list of highly correlated variables with more than 90% of correlation*

'Curr_Assets', 'Curr_Liab_and_Prov', 'Total_Assets_to_Liab', 'Net_Sales', 'Value_Of_Output', 'Cost_of_Prod', 'PBIT', 'PBT', 'PAT', 'Adjusted_PAT', 'CP', 'ROG_Net_Sales_perc', 'ROG_PBDT_perc', 'ROG_PBIT_perc', 'ROG_PAT_perc', 'ROG_CP_perc', 'CPM_perc_Latest', 'APATM_perc_Latest'

# 4.2 Correlation Plot



After dimensionality reduction
Shape of dataset: (3586, 44)

Reduced from 65 to 44

## 5.1 Target Variable & Scaling

- Output variable [y= default] is separated from the dataset which is renamed as X after separation.
- Now, scaling is required before proceeding with any model building as different variables are at different scale.
- Scaling was done using standard scaler. Glimpse of scaled dataset is as shown below.

| | Equity_Paid_Up | Networth | Capital_Employed | Total_Debt | Gross_Block | Net_Working_Capital | Gross_Sales | Other_Income | Selling_Cost | PBIDT | PBDT |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2.08394 | -2.124536 | -2.277679 | 1.955653 | 2.011189 | -2.126873 | -0.711425 | 1.501295 | -0.666143 | -2.077966 | -1.859258 |
| 1 | 2.08394 | -2.124536 | 1.937167 | 1.955653 | 2.011189 | -2.126873 | 1.980032 | 1.924923 | 1.942957 | 1.877612 | -0.797237 |
| 2 | 2.08394 | 1.892586 | 1.937167 | 1.955653 | 2.011189 | 1.933358 | 1.037488 | 1.924923 | 1.942957 | -2.077966 | -1.859258 |
| 3 | 2.08394 | -2.124536 | 1.937167 | 1.955653 | 2.011189 | -2.126873 | 1.980032 | 1.924923 | 0.231669 | -2.077966 | -1.859258 |
| 4 | 2.08394 | -2.124536 | 1.937167 | 1.955653 | 2.011189 | 1.933358 | -0.538733 | 1.924923 | -0.136595 | -2.077966 | -1.859258 |

- Dataset is cleaned from any NaN data.
- Train & Test dataset is created with ratio of 67:33 & random-state=42.
- Train has 43 columns and 2402 rows.
- Test has 43 columns and 1184 rows.
- 43 variables are a huge and can mess up our prediction. Hence we will do initial feature selection using recursive feature elimination and pick the best 15 features among rest.

# 5.2 Recursive Feature Elimination using LR

| | Feature | Rank |
|---|---|---|
| 0 | Equity_Paid_Up | 28 |
| 1 | Networth | 1 |
| 2 | Capital_Employed | 1 |
| 3 | Total_Debt | 1 |
| 4 | Gross_Block | 20 |
| 5 | Net_Working_Capital | 4 |
| 6 | Gross_Sales | 10 |
| 7 | Other_Income | 8 |
| 8 | Selling_Cost | 1 |
| 9 | PBIDT | 1 |
| 10 | PBDT | 18 |
| 11 | Rev_earn_in_forex | 13 |
| 12 | Rev_exp_in_forex | 1 |
| 13 | Book_Value_Unit_Curr | 1 |
| 14 | Book_Value_Adj_Unit_Curr | 29 |
| 15 | Market_Capitalisation | 1 |
| 16 | CEPS_annualised_Unit_Curr | 1 |
| 17 | Cash_Flow_From_Opr | 26 |
| 18 | Cash_Flow_From_Inv | 14 |
| 19 | Cash_Flow_From_Fin | 27 |
| 20 | ROG_Net_Worth_perc | 1 |
| 21 | ROG_Capital_Employed_perc | 6 |
| 22 | ROG_Gross_Block_perc | 12 |
| 23 | ROG_Gross_Sales_perc | 19 |
| 24 | ROG_Cost_of_Prod_perc | 2 |
| 25 | ROG_Total_Assets_perc | 5 |
| 26 | ROG_PBIDT_perc | 24 |
| 27 | ROG_PBT_perc | 15 |
| 28 | ROG_Market_Capitalisation_perc | 21 |
| 29 | Curr_Ratio_Latest | 1 |
| 30 | Fixed_Assets_Ratio_Latest | 3 |
| 31 | Inventory_Ratio_Latest | 1 |
| 32 | Debtors_Ratio_Latest | 1 |
| 33 | Total_Asset_Turnover_Ratio_Latest | 17 |
| 34 | Interest_Cover_Ratio_Latest | 1 |
| 35 | PBIDTM_perc_Latest | 23 |
| 36 | PBITM_perc_Latest | 22 |
| 37 | PBDTM_perc_Latest | 9 |
| 38 | Debtors_Vel_Days | 7 |
| 39 | Creditors_Vel_Days | 16 |
| 40 | Inventory_Vel_Days | 25 |
| 41 | Value_of_Output_to_Total_Assets | 11 |
| 42 | Value_of_Output_to_Gross_Block | 1 |

Selected top 15 features by recursive elimination using Logistic Regression

| | Feature | Rank |
|---|---|---|
| 1 | Networth | 1 |
| 2 | Capital_Employed | 1 |
| 3 | Total_Debt | 1 |
| 8 | Selling_Cost | 1 |
| 9 | PBIDT | 1 |
| 12 | Rev_exp_in_forex | 1 |
| 13 | Book_Value_Unit_Curr | 1 |
| 15 | Market_Capitalisation | 1 |
| 16 | CEPS_annualised_Unit_Curr | 1 |
| 20 | ROG_Net_Worth_perc | 1 |
| 29 | Curr_Ratio_Latest | 1 |
| 31 | Inventory_Ratio_Latest | 1 |
| 32 | Debtors_Ratio_Latest | 1 |
| 34 | Interest_Cover_Ratio_Latest | 1 |
| 42 | Value_of_Output_to_Gross_Block | 1 |

# 5.3 Modelling using Logistic Regression & Feature Selection

# Modelling Results from Model_1 started with 15 features

Logit Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | default | **No. Observations:** | 2402 |
| **Model:** | Logit | **Df Residuals:** | 2386 |
| **Method:** | MLE | **Df Model:** | 15 |
| **Date:** | Mon, 06 Dec 2021 | **Pseudo R-squ.:** | 0.5980 |
| **Time:** | 11:40:46 | **Log-Likelihood:** | -331.01 |
| **converged:** | True | **LL-Null:** | -823.47 |
| **Covariance Type:** | nonrobust | **LLR p-value:** | 2.301e-200 |

| | coef | std err | z | P>\|z\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | -7.5241 | 0.505 | -14.887 | 0.000 | -8.515 | -6.533 |
| **Networth** | -0.6488 | 0.417 | -1.557 | 0.119 | -1.465 | 0.168 |
| **Capital_Employed** | -0.5778 | 0.566 | -1.020 | 0.308 | -1.688 | 0.532 |
| **Total_Debt** | 1.3585 | 0.375 | 3.622 | 0.000 | 0.623 | 2.094 |
| **Selling_Cost** | -0.3243 | 0.278 | -1.168 | 0.243 | -0.868 | 0.220 |
| **PBIDT** | -0.5765 | 0.329 | -1.752 | 0.080 | -1.221 | 0.068 |
| **Rev_exp_in_forex** | 0.3099 | 0.226 | 1.372 | 0.170 | -0.133 | 0.753 |
| **Book_Value_Unit_Curr** | -6.0852 | 0.643 | -9.460 | 0.000 | -7.346 | -4.824 |
| **Market_Capitalisation** | -0.5763 | 0.307 | -1.880 | 0.060 | -1.177 | 0.024 |
| **CEPS_annualised_Unit_Curr** | -0.4984 | 0.354 | -1.406 | 0.160 | -1.193 | 0.196 |
| **ROG_Net_Worth_perc** | -0.4011 | 0.132 | -3.045 | 0.002 | -0.659 | -0.143 |
| **Curr_Ratio_Latest** | -0.6999 | 0.651 | -1.074 | 0.283 | -1.977 | 0.577 |
| **Inventory_Ratio_Latest** | -1.5284 | 1.127 | -1.356 | 0.175 | -3.738 | 0.681 |
| **Debtors_Ratio_Latest** | -1.1137 | 1.821 | -0.612 | 0.541 | -4.683 | 2.455 |
| **Interest_Cover_Ratio_Latest** | -0.4314 | 0.329 | -1.312 | 0.190 | -1.076 | 0.213 |
| **Value_of_Output_to_Gross_Block** | -0.4849 | 0.160 | -3.027 | 0.002 | -0.799 | -0.171 |

There are few insignificant variables that need to be eliminated further.

# 5.3.1 VIF for Multicollinearity check

# using VIF for multi-collinearity check

| | variables | VIF |
|---|---|---|
| 1 | Capital_Employed | 10.551030 |
| 0 | Networth | 7.834030 |
| 4 | PBIDT | 4.958160 |
| 8 | CEPS_annualised_Unit_Curr | 3.867861 |
| 2 | Total_Debt | 3.726908 |
| 6 | Book_Value_Unit_Curr | 2.883421 |
| 7 | Market_Capitalisation | 2.637396 |
| 3 | Selling_Cost | 2.561793 |
| 5 | Rev_exp_in_forex | 2.035738 |
| 9 | ROG_Net_Worth_perc | 1.669261 |
| 14 | Value_of_Output_to_Gross_Block | 1.119559 |
| 13 | Interest_Cover_Ratio_Latest | 1.059009 |
| 12 | Debtors_Ratio_Latest | 1.013748 |
| 11 | Inventory_Ratio_Latest | 1.013310 |
| 10 | Curr_Ratio_Latest | 1.007535 |

Variance inflation factor (VIF) is a measure of the amount of multicollinearity in a set of multiple regression variables.

VIF should be below 5.
'Capital_Employed' has the highest VIF value above 7, hence should be eliminated and then rechecked.



| | variables | VIF |
|---|---|---|
| 3 | PBIDT | 4.937241 |
| 0 | Networth | 4.186332 |
| 7 | CEPS_annualised_Unit_Curr | 3.863714 |
| 5 | Book_Value_Unit_Curr | 2.882601 |
| 6 | Market_Capitalisation | 2.620175 |
| 2 | Selling_Cost | 2.558161 |
| 1 | Total_Debt | 2.056565 |
| 4 | Rev_exp_in_forex | 2.026963 |
| 8 | ROG_Net_Worth_perc | 1.652877 |
| 13 | Value_of_Output_to_Gross_Block | 1.118411 |
| 12 | Interest_Cover_Ratio_Latest | 1.049056 |
| 11 | Debtors_Ratio_Latest | 1.013583 |
| 10 | Inventory_Ratio_Latest | 1.013265 |
| 9 | Curr_Ratio_Latest | 1.007420 |

Now all the VIF values are under 5 and can be proceeded.

# 5.3.2 Modelling using Logistic Regression & Feature Selection

# Modelling Results from Model_8

Logit Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | default | **No. Observations:** | 2402 |
| **Model:** | Logit | **Df Residuals:** | 2393 |
| **Method:** | MLE | **Df Model:** | 8 |
| **Date:** | Mon, 06 Dec 2021 | **Pseudo R-squ.:** | 0.5926 |
| **Time:** | 12:04:09 | **Log-Likelihood:** | -335.46 |
| **converged:** | True | **LL-Null:** | -823.47 |
| **Covariance Type:** | nonrobust | **LLR p-value:** | 2.242e-205 |

| | coef | std err | z | P>\|z\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | -7.3385 | 0.483 | -15.186 | 0.000 | -8.286 | -6.391 |
| **Networth** | -0.8121 | 0.354 | -2.297 | 0.022 | -1.505 | -0.119 |
| **Total_Debt** | 1.1298 | 0.199 | 5.677 | 0.000 | 0.740 | 1.520 |
| **PBIDT** | -0.7743 | 0.286 | -2.712 | 0.007 | -1.334 | -0.215 |
| **Book_Value_Unit_Curr** | -6.2276 | 0.640 | -9.732 | 0.000 | -7.482 | -4.973 |
| **Market_Capitalisation** | -0.7470 | 0.286 | -2.608 | 0.009 | -1.308 | -0.186 |
| **ROG_Net_Worth_perc** | -0.4691 | 0.122 | -3.852 | 0.000 | -0.708 | -0.230 |
| **Inventory_Ratio_Latest** | -1.9615 | 1.283 | -1.529 | 0.126 | -4.475 | 0.552 |
| **Value_of_Output_to_Gross_Block** | -0.4720 | 0.158 | -2.990 | 0.003 | -0.781 | -0.163 |

**Features eliminated:**

Capital_Employed >
Debtors_Ratio_Latest >
Curr_Ratio_Latest >
Selling_Cost >
Rev_exp_in_forex >
CEPS_annualised_Unit_Curr >
Interest_Cover_Ratio_Latest

There are still one insignificant variable (Inventory_Ratio_Latest) that needs to be eliminated.

## 5.3.4 Final Model after Feature Selection

# Modelling Results from Model_9

Logit Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | default | **No. Observations:** | 2402 |
| **Model:** | Logit | **Df Residuals:** | 2394 |
| **Method:** | MLE | **Df Model:** | 7 |
| **Date:** | Mon, 06 Dec 2021 | **Pseudo R-squ.:** | 0.5914 |
| **Time:** | 12:05:31 | **Log-Likelihood:** | -336.44 |
| **converged:** | True | **LL-Null:** | -823.47 |
| **Covariance Type:** | nonrobust | **LLR p-value:** | 4.858e-206 |

| | coef | std err | z | P>\|z\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | -7.2051 | 0.466 | -15.468 | 0.000 | -8.118 | -6.292 |
| **Networth** | -0.7906 | 0.351 | -2.250 | 0.024 | -1.479 | -0.102 |
| **Total_Debt** | 1.1148 | 0.198 | 5.620 | 0.000 | 0.726 | 1.504 |
| **PBIDT** | -0.7480 | 0.285 | -2.625 | 0.009 | -1.306 | -0.190 |
| **Book_Value_Unit_Curr** | -6.2362 | 0.639 | -9.761 | 0.000 | -7.488 | -4.984 |
| **Market_Capitalisation** | -0.7612 | 0.284 | -2.680 | 0.007 | -1.318 | -0.205 |
| **ROG_Net_Worth_perc** | -0.4728 | 0.121 | -3.894 | 0.000 | -0.711 | -0.235 |
| **Value_of_Output_to_Gross_Block** | -0.4841 | 0.157 | -3.080 | 0.002 | -0.792 | -0.176 |

All the p values are now in acceptable range. Hence, we can finalise here with model 9 and proceed further with the prediction and validation.

## 6.1 Prediction

# Prediction on Train & Test dataset

Predicted Train dataset

| | |
|---|---|
| 842 | 0.0 |
| 1057 | 0.0 |
| 1595 | 0.0 |
| 100 | 1.0 |
| 1191 | 0.0 |
| 2163 | 0.0 |
| 2763 | 0.0 |
| 2701 | 0.0 |
| 2072 | 0.0 |
| 2349 | 0.0 |
| 1392 | 0.0 |
| 1621 | 0.0 |
| 1960 | 0.0 |
| 2148 | 0.0 |
| 571 | 0.0 |
| 1984 | 0.0 |
| 1592 | 0.0 |
| 3110 | 0.0 |
| 1564 | 0.0 |

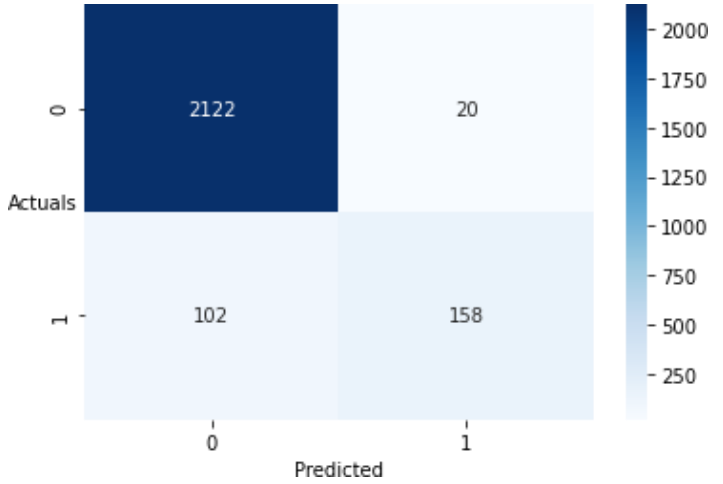
Predicted Test dataset

| | |
|---|---|
| 251 | 1.0 |
| 3493 | 0.0 |
| 3063 | 0.0 |
| 2384 | 0.0 |
| 1679 | 0.0 |
| 604 | 0.0 |
| 3434 | 0.0 |
| 2244 | 0.0 |
| 2523 | 0.0 |
| 2162 | 0.0 |
| 3102 | 0.0 |
| 1638 | 0.0 |
| 2046 | 0.0 |
| 1241 | 0.0 |
| 133 | 0.0 |
| 2294 | 0.0 |
| 2139 | 0.0 |
| 2844 | 0.0 |
| 1360 | 0.0 |

## 6.2.1 Validation (Thres=0.5)

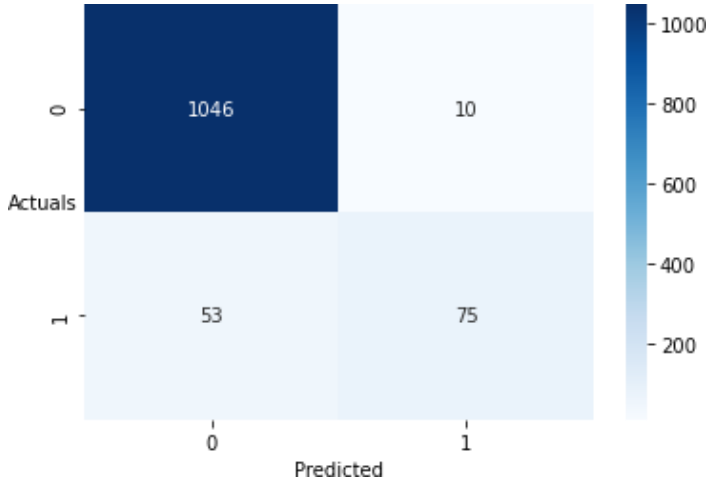
# Confusion Matrix & Classification Report for Train & Test dataset (with threshold=0.5)

### Predicted Train dataset




| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.954 | 0.991 | 0.972 | 2142 |
| 1.0 | 0.888 | 0.608 | 0.721 | 260 |
| accuracy | | | 0.949 | 2402 |
| macro avg | 0.921 | 0.799 | 0.847 | 2402 |
| weighted avg | 0.947 | 0.949 | 0.945 | 2402 |

### Predicted Test dataset



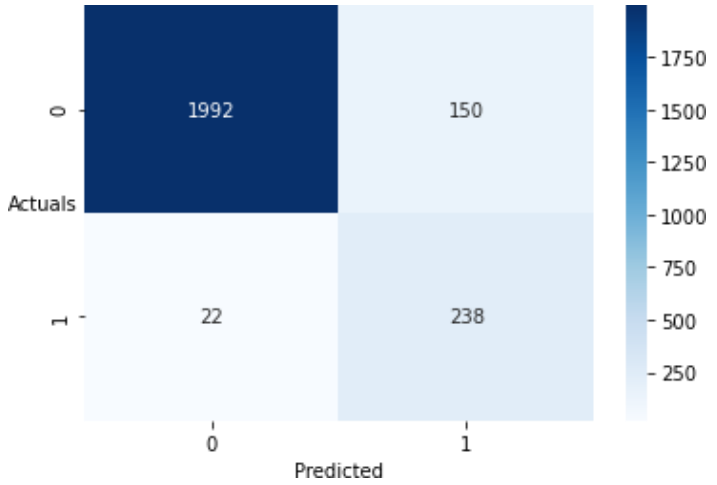

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.952 | 0.991 | 0.971 | 1056 |
| 1.0 | 0.882 | 0.586 | 0.704 | 128 |
| accuracy | | | 0.947 | 1184 |
| macro avg | 0.917 | 0.788 | 0.837 | 1184 |
| weighted avg | 0.944 | 0.947 | 0.942 | 1184 |

Precision to measure default on test dataset is 88% with recall of 58% and accuracy of 94%.

## 6.2.2 Validation (Thres= 0.168)

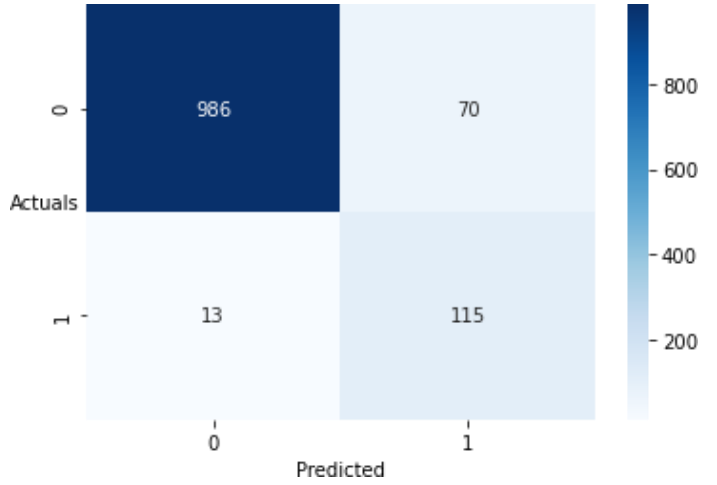
# Confusion Matrix & Classification Report for Train & Test dataset (with threshold=0.1689)

### Predicted Train dataset




| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.989 | 0.930 | 0.959 | 2142 |
| 1.0 | 0.613 | 0.915 | 0.735 | 260 |
| accuracy | | | 0.928 | 2402 |
| macro avg | 0.801 | 0.923 | 0.847 | 2402 |
| weighted avg | 0.948 | 0.928 | 0.934 | 2402 |

### Predicted Test dataset



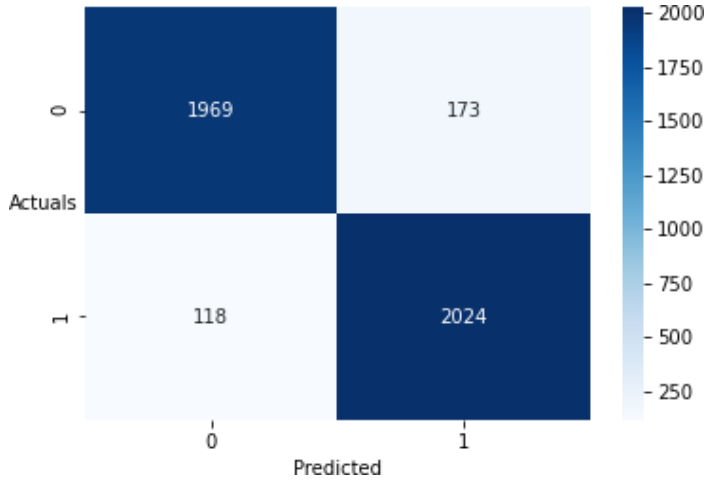

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.987 | 0.934 | 0.960 | 1056 |
| 1.0 | 0.622 | 0.898 | 0.735 | 128 |
| accuracy | | | 0.930 | 1184 |
| macro avg | 0.804 | 0.916 | 0.847 | 1184 |
| weighted avg | 0.947 | 0.930 | 0.935 | 1184 |

Precision to measure default on test dataset is 62% with recall of 90% and accuracy of 93%.

## 6.2.3 Validation (SMOTE)

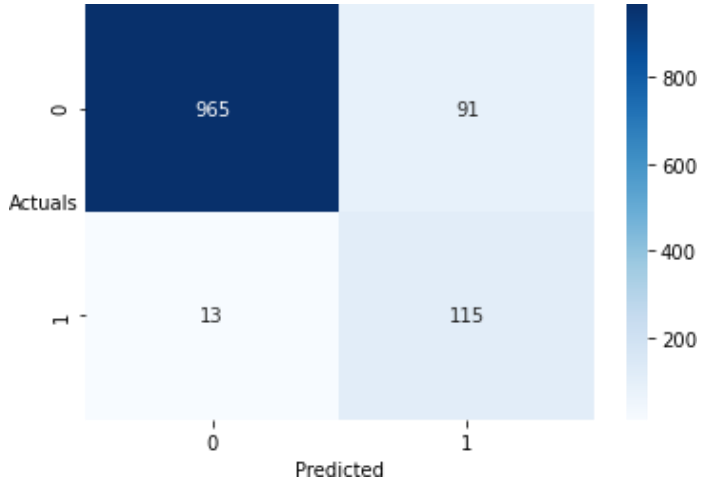
# Confusion Matrix & Classification Report for Train & Test dataset (using SMOTE)

### Predicted Train dataset




| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.94 | 0.92 | 0.93 | 2142 |
| 1.0 | 0.92 | 0.94 | 0.93 | 2142 |
| accuracy | | | 0.93 | 4284 |
| macro avg | 0.93 | 0.93 | 0.93 | 4284 |
| weighted avg | 0.93 | 0.93 | 0.93 | 4284 |

### Predicted Test dataset




| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.99 | 0.91 | 0.95 | 1056 |
| 1.0 | 0.56 | 0.90 | 0.69 | 128 |
| accuracy | | | 0.91 | 1184 |
| macro avg | 0.77 | 0.91 | 0.82 | 1184 |
| weighted avg | 0.94 | 0.91 | 0.92 | 1184 |

Precision to measure default on test dataset is 56% with recall of 90% and accuracy of 91%.

# # CONCLUSION

- The dataset was huge, hence clean and processed into a neat dataset.
- The outliers were treated.
- Univariate and Bivariate analysis was done to check the data behaviour and any early sign of multi-collinearity.
- We used various feature selection and elimination method to remove huge number of redundant variables.
- We used Logistic Regression for modelling and taken two different threshold values to predict the output. One threshold was manually set and another was calculated.
- Predictions were made based on the final model.
- We also used SMOTE for the betterment of model.
- For ref. Jupyter Notebook file is included.